

CNC Reference

Safety Block: G40 G17 G80 G90

Cancels Comp (G40), Selects XY (G17), Cancels Cycles (G80), Sets Absolute (G90).

1. G-Codes (V=VMC, L=Lathe, S=Swiss, E=EDM)

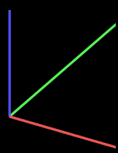
Code	Name	V	L	S	E
G00	Rapid Position	✓	✓	✓	✓
G01	Linear Interp	✓	✓	✓	✓
G02	CW Arc	✓	✓	✓	✓
G03	CCW Arc	✓	✓	✓	✓
G04	Dwell	✓	✓	✓	✓
G10	Data Setting (Fanuc)	✓	✓	✓	
G17	XY Plane	✓		✓	✓
G18	XZ Plane	✓	✓	✓	
G19	YZ Plane	✓		✓	✓
G20	Imperial	✓	✓	✓	✓
G21	Metric	✓	✓	✓	✓
G28	Return Home	✓	✓	✓	✓
G31	Skip / Probe (Fanuc)	✓	✓	✓	
G40	Comp Cancel	✓	✓	✓	✓
G41	Comp Left	✓	✓	✓	✓
G42	Comp Right	✓	✓	✓	✓
G43	Length Comp	✓		✓	
G54	Work Offsets	✓	✓	✓	✓
G54.1	Extended Offsets (P1-P48)	✓	✓		
G61	Exact Stop	✓	✓	✓	✓
G65	Macro Call (Fanuc)	✓	✓	✓	
G71	Rough Turn		✓	✓	

G76	Threading		✓	✓	
G80	Cycle Cancel	✓	✓	✓	✓
G81	Drill Cycle	✓	✓	✓	
G83	Peck Drill	✓	✓	✓	
G90	Absolute	✓	✓	✓	✓
G91	Incremental	✓	✓	✓	✓
G96	Constant Surf Spd		✓	✓	
G97	Constant RPM	✓	✓	✓	
G112	Polar Interp		✓	✓	
G300	Swiss Sync			✓	

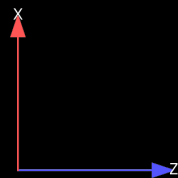
2. M-Codes

Code	Name	V	L	S	E
M00	Prog Stop	✓	✓	✓	✓
M01	Opt Stop	✓	✓	✓	✓
M03	Spindle CW	✓	✓	✓	
M04	Spindle CCW	✓	✓	✓	
M05	Spindle Stop	✓	✓	✓	
M06	Tool Change	✓			
M08	Coolant ON	✓	✓	✓	
M09	Coolant OFF	✓	✓	✓	
M20	EDM Flush ON				✓
M30	Prog End	✓	✓	✓	✓
M50	Wire Cut				✓
M60	Wire Thread				✓
M98	Sub Call (Fanuc)	✓	✓	✓	✓
M99	Sub Return (Fanuc)	✓	✓	✓	✓
M200	Wait/Sync			✓	

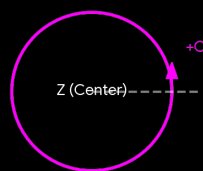
3. Coordinate Diagrams



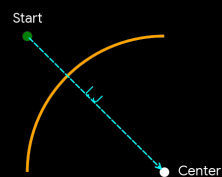
Mill (3-Axis)



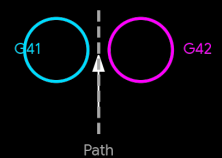
Lathe (2-Axis)



C-Angle



I,J,K Arcs



G41/G42

4. Sample Programs (Fanuc Standard)

(VMC Example with Fanuc Extended Offsets)

```
O1001                (Program Name)
G21 G17 G90 G40 G80  (Safety)
G54.1 P1 T1 M06      (Ext Offset 1, Tool Change)
S1500 M03            (1500 RPM, Fwd)
G00 X0 Y0            (Rapid Move)
G43 H01 Z50.0         (Length Comp)
G01 Z-5.0 F200        (Feed to depth)
M30                  (End)
```

Controller Ecosystem

Fanuc (Industry Standard)

Architecture: Highly reliable, standardized ISO G-code. Dominates the Asian and American markets.

Key Features: Uses *Custom Macro B* (G65) for parametric programming. Strict formatting. Requires exact syntax (#100=1).

Common On: Doosan, Matsuura, Makino, Mori Seiki.

Siemens (Sinumerik)

Architecture: Highly advanced, PC-based. Dominates the European market.

Key Features: *ShopMill* & *ShopTurn* conversational programming is built-in. High-level language native support (e.g., IF/THEN/ELSE without strict macro calls). Variables are named (e.g., R1=10).

Common On: DMG Mori, Grob, Hermle.

Mitsubishi (Meldas / M800)

Architecture: Extremely fast processing, G-code syntax is highly compatible with Fanuc.

Key Features: Known for advanced high-speed look-ahead for complex 3D mold surfacing. Excellent *Navi* graphical interfaces.

Common On: Mazak (customized as Mazatrol), Citizen (Swiss).

Heidenhain (TNC)

Architecture: Dialog-based conversational programming (Klartext).

Key Features: Does not rely on standard G-code (though it can run ISO). Uses distinct buttons for lines (L), circles (C), and chamfers (CHF). Superior for 5-axis mold making.

Common On: Hermle, Mikron, Kern.

Haas (Next Generation Control)

Architecture: User-friendly, proprietary but heavily Fanuc-based.

Key Features: Visual Programming System (VPS), highly intuitive tool setup screens, specific M-codes (M130 for media display). Incredible documentation.

Common On: Haas Automation exclusively.